Category Theory in Mathlib

ScottLean4 · for Dana Scott

Grounded in the copy on disk (Lean 4 `v4.32.2`): `Mathlib/CategoryTheory/` is about **1,089 files** — one of the largest formalized category-theory libraries anywhere. Its shape:

Foundations

- `Category` (the class): objects in `Type u`, hom-types `X □ Y` in `Type v` (universe-polymorphic, so small vs. large is handled by universes), identity `□ X`, composition written *diagrammatically* `f □ g` ("`f` then `g`"), and the three laws. A `Category` extends `Quiver` (a graph) — quivers are the substrate.

- `Functor` (`C □ D`), `NatTrans` / `NatIso`, `Iso`, `Equivalence` of categories, and **Yoneda** (embedding and lemma).

Universal constructions — the bulk

- `Limits` — *208 files*: cones/cocones, all (co)limits, products/coproducts, equalizers, pullbacks/pushouts, terminal/initial objects, **Kan extensions**, filtered colimits, and the full preservation/reflection/creation apparatus.

- `Adjunction` (22): units/counits, reflective subcategories, adjoint-functor theorems, monadicity.

- `Comma` (17), `ComposableArrows`, over/under (slice) categories, `WithTerminal`.

Monoidal, enriched, higher

- `Monoidal` — *108 files*: monoidal, braided, symmetric, **closed** and **cartesian-closed**, rigid/autonomous categories, coherence, monoidal functors. (Mathlib's CCC lives here.)

- `Enriched` (10), `Bicategory` (44: 2-categories, pseudofunctors), `Center`.

The homological tower

`Preadditive` (34) → `Linear` → `Abelian` (56: kernels/cokernels, exactness) → `Triangulated` (30) with `Shift` (17) → `Localization` (40: calculus of fractions, the machinery behind derived categories). Also `Idempotents` (Karoubi envelope), `GradedObject`, `Subobject`.

Topos-theoretic / geometric

`Sites` — *113 files*: Grothendieck topologies, sheaves on a site, sheafification. Plus `Topos` (elementary topoi), `Galois` (11: Galois categories & the fundamental group), `LocallyCartesianClosed`, `Distributive`, `EffectiveEpi`, `RegularCategory`, `Adhesive`.

Monads & special categories

`Monad` (12): monads, **Eilenberg–Moore** and **Kleisli** categories, `Endofunctor`. And `Groupoid`, `Discrete`, `FinCategory`, `Filtered`, `Types` (the category of Lean types), `ConcreteCategory` (a faithful functor to `Type` — forgetful functors), `Quotient`, `PathCategory` (free category on a quiver).

The everyday categories (filed elsewhere)

Concrete categories of structures live under their subject, not `CategoryTheory/`: `Algebra/Category/` (`MonCat`, `GrpCat`, `RingCat`, `ModuleCat`, …), `Order/Category/` (`Lat`, `DistLat`, `HeytAlg`, `Preord`, …), `Topology/Category/` (`TopCat`, `Profinite`, …) — each with (co)completeness and forgetful/free adjunctions proved.

Two design points

- **Automation:** a dedicated tactic `aesop_cat` discharges the naturality/coherence side-goals, so most bookkeeping is automated; `simp` carries the rewriting.
- **Modern frontier:** `Presentable`/accessible categories (18), the small-object argument (`SmallObject`), and `MorphismProperty`/`ObjectProperty` (classes of maps/objects — the model-category-flavored infrastructure), `LiftingProperties`.

A few in your own orbit

`Topos`, `Galois` categories, and cartesian-/locally-cartesian-closed structure are all present; there is even a `Dialectica` category (Gödel's Dialectica, categorically) and `MarkovCategory`/`CopyDiscardCategory` (categorical probability, à la Fritz).